

PHARMACY DEVELOPER GUIDE

Database Architecture, Models & Security Rules (डेवलपर गाइड)

Note for Developers: This section contains database relationships, Laravel model architectures, FEFO logic scopes, and SQL security guidelines.

1. Database Schema & Architecture (डेटाबेस संरचना)

Table Name	Description & Mapping Logic
pharmacy_suppliers	Stores supplier/vendor details. - party_type = 1,2,3: Customer & Supplier details - party_type = 2,4: Vendor details only
pharmacy_categories	Master category table (e.g., Tablets, Syrups, Injections).
pharmacy_medicines	Master drugs list mapped with customer_id, supplier_id, and category_id.
pharmacy_medicine_batches	Tracks stock quantities, batch details, pricing, GST, and expiry dates.

2. Multi-Tenant Security & Rules (मल्टी-टेनेंट सुरक्षा)

A> Tenant Isolation:
Every database operation MUST strictly filter by customer_id.

```
// Always apply customer context in queries
$medicines = PharmacyMedicine::where('customer_id', authUser()->customer_id)->get();
```

B> Column Ambiguity Protection:
When using Joins, always prefix column names with table aliases.

```
DB::table('pharmacy_medicines')
->join('pharmacy_categories', 'pharmacy_medicines.category_id', '=', 'pharmacy_categories.id')
->where('pharmacy_medicines.customer_id', authUser()->customer_id)
->select('pharmacy_medicines.*', 'pharmacy_categories.name as category_name');
```

3. Laravel Eloquent Models & Scopes

```
// 1. PharmacyMedicine.php
class PharmacyMedicine extends Model {
    protected $table = 'pharmacy_medicines';

    public function batches() {
        return $this->hasMany(PharmacyMedicineBatch::class, 'medicine_id');
    }

    public function category() {
```

```
        return $this->belongsTo(PharmacyCategory::class, 'category_id');
    }
}

// 2. PharmacyMedicineBatch.php (FEFO Scope Implementation)
class PharmacyMedicineBatch extends Model {
    protected $table = 'pharmacy_medicine_batches';

    // Auto FEFO Scope: First Expire, First Out
    public function scopeAvailableFefo($query) {
        return $query->where('expiry_date', '>', now())
            ->where('purchase_quantity', '>', 0)
            ->orderBy('expiry_date', 'asc');
    }
}
```